Why Program?

Chapter 1

Python for Everybody

www.py4e.com

UMSI

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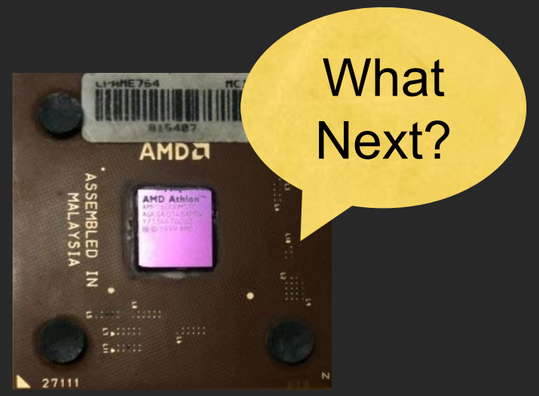
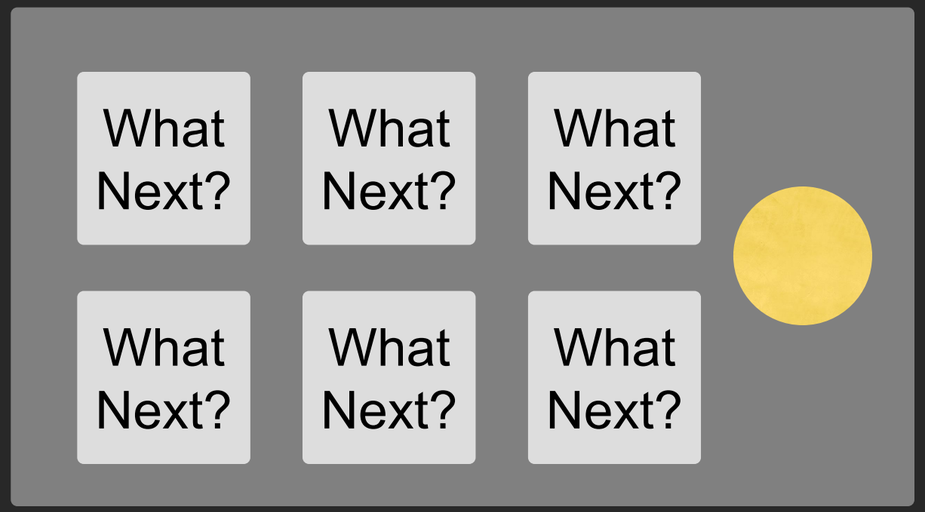
Computers want to be helpful...

- Computers are built for one purpose - to do things for us
- But we need to speak their language to describe what we want done
- Users have it easy - someone already put many different programs (instructions) into the computer and users just pick the ones they want to use





Programmers Anticipate Needs

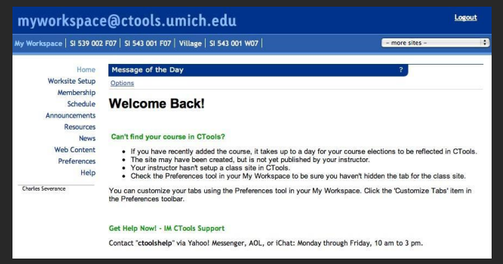
- iPhone Applications are a market
- iPhone Applications have over 3 Billion downloads
- Programmers have left their jobs to be full-time iPhone developers
- Programmers know the ways of the program

Users vs. Programmers

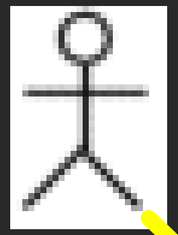
- Users see computers as a set of tools - word processor, spreadsheet, map, to-do list, etc.
- Programmers learn the computer “ways” and the computer language
- Programmers have some tools that allow them to build new tools
- Programmers sometimes write tools for lots of users and sometimes programmers write little “helpers” for themselves to automate a task

Why be a programmer?

- To get some task done - we are the user and programmer
  - Clean up survey data
- To produce something for others to use - a programming job
  - Fix a performance problem in the Sakai software
  - Add a guestbook to a web site



User



Computer
Hardware + Software

Programmer

Data

Information

....

Networks

From a software creator's point of view, we build the software. The end users (stakeholders/actors) are our masters - who we want to please - often they pay us money when they are pleased. But the data, information, and networks are our problem to solve on their behalf. The hardware and software are our friends and allies in this quest.

What is Code? Software? A Program?

- A sequence of stored instructions
  - It is a little piece of our intelligence in the computer
  - We figure something out and then we encode it and then give it to someone else to save them the time and energy of figuring it out
- A piece of creative art - particularly when we do a good job on user experience

Programs for Humans...

https://www.youtube.com/watch?v=gwWRjvwlLKg

Programs for Humans...

```
while music is playing:
    Left hand out and up
    Right hand out and up
    Flip Left hand
    Flip Right hand
    Left hand to right shoulder
    Right hand to left shoulder
    Left hand to back of head
    Right ham to back of head
    Left hand to right hit
    Right hand to left hit
    Left hand on left bottom
    Right hand on right bottom
    Wiggle
    Wiggle
    Jump
```

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    Right hand to left hip
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    Wiggle
    Wiggle
    Jump
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Programs for Python...

the clown ran after the car and the car ran into the tent and the tent fell down on the clown and the car

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Programs for Python...

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```
name = input('Enter file:')
handle = open(name)

counts = dict()
for line in handle:
    words = line.split()
    for word in words:
        counts[word] = counts.get(word,0) + 1

bigcount = None
bigword = None
for word,count in counts.items():
    if bigcount is None or count > bigcount:
        bigword = word
        bigcount = count

print(bigword, bigcount)
```

```
python words.py
Enter file: words.txt
to 16
```

```
python words.py
Enter file: clown.txt
the 7
```

Hardware Architecture